

# Full Stack Development with MERN

## Project Documentation format

### 1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:** Team 2 – members and roles:

| Team Member    | Role                                          |
|----------------|-----------------------------------------------|
| Akshaya E      | Service Catalog item, request form and portal |
| Akshaya S      | Approval configuration and testing            |
| Akshaya L      | Flow Designer flow and procurement tasks      |
| Allin Kiruba R | Notifications, asset update and documentation |

### 2. Project Overview

- **Purpose:** This project demonstrates how to automate the standard laptop ordering process using ServiceNow Flow Designer. Employees submit laptop requests through the Service Catalog, and the workflow automatically processes the request, manages approvals when required, creates procurement tasks and sends notifications. The goal is to reduce manual effort, minimize errors and improve the efficiency of IT procurement management.
- **Features:**
  - o Service Catalog item for ordering a standard laptop
  - o Automatic flow trigger when a request is submitted
  - o Conditional manager approval (approve / reject with comments)
  - o Automatic creation of a procurement task for the IT procurement group
  - o Email notifications at each stage (submission, decision, delivery)
  - o Request status tracking from the portal and execution logs in Flow Designer

### 3. Architecture

- **Frontend:** The user interface is the ServiceNow Service Portal. Employees use the Service Catalog item and its request form (built with form variables, catalog client scripts and UI policies); portal widgets (AngularJS, HTML, CSS) display the catalog and My Requests pages.
- **Backend:** The backend runs on the ServiceNow platform. A Flow Designer flow is triggered by the Service Catalog request and runs the actions: check conditions, Ask for Approval, Create Catalog Task, and Send Email. Server-side logic uses Flow Designer actions with JavaScript where needed.
- **Database:** Data is stored in the ServiceNow relational database and accessed through its tables:
  - o sc\_request / sc\_req\_item – the request and requested item
  - o sysapproval\_approver – manager approval records
  - o sc\_task – procurement task assigned to the IT procurement group
  - o sys\_user / sys\_user\_group – requesters, managers and groups
  - o alm\_hardware (CMDB) – laptop asset records

### 4. Setup Instructions

- **Prerequisites:**
  - o ServiceNow Personal Developer Instance (PDI) with admin access
  - o Web browser (Chrome / Edge) with internet access
  - o Users with a manager assigned, and an 'IT Procurement' group with members
  - o Outbound email configured on the instance for notifications
- **Installation:**
  1. Sign in to the ServiceNow instance as an administrator.

2. Create the catalog item 'Standard Laptop Request' with variables (laptop model, business justification, delivery location) and publish it to the Service Catalog.
3. Create the group 'IT Procurement' and make sure requesters have a manager set on their user record.
4. Open Flow Designer and create a new flow with the trigger 'Service Catalog'.
5. Add the actions: Ask for Approval (requester's manager), Create Catalog Task (assign to IT Procurement), and Send Email notifications.
6. Save and activate the flow, then test it from the catalog item.

## 5. Folder Structure

ServiceNow stores configuration records inside the instance (and application scope) instead of folders. The equivalent structure is:

- **Client:** Service Portal page and widgets; Service Catalog item (form variables); Catalog Client Scripts and UI Policies for the request form.
- **Server:** Flow Designer flow and its actions; Business Rules / Script Includes (if used); Email Notifications; Catalog Tasks; Access Control rules (ACLs).

## 6. Running the Application

*Provide commands to start the frontend and backend servers locally.*

- **Frontend:** No local server is needed. Open the instance Service Portal in a browser (<https://<your-instance>.service-now.com/sp>) and open the Standard Laptop Request item.
- **Backend:** No local server is needed. The flow runs on the instance once it is activated; use the Test option in Flow Designer and monitor each run under Flow Designer > Executions.

## 7. API Documentation

The platform exposes REST APIs that can be used to place and track laptop requests. Example endpoints (replace the instance name and sys\_id values):

| Method | Endpoint                                           | Parameters                                                         | Example response                                                     |
|--------|----------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------|
| POST   | /api/sn_sc/servicecatalog/items/{sys_id}/order_now | sysparm_quantity,<br>variables (model,<br>justification, location) | {"result":{"number":"REQ0010001","request_id":"..."}}                |
| GET    | /api/now/table/sc_req_item                         | sysparm_query (e.g.<br>request=REQ0010001),<br>sysparm_limit       | {"result":[{"number":"RITM0010001","stage":"Waiting for Approval"}]} |
| GET    | /api/now/table/sysapproval_approver                | sysparm_query<br>(document_id=<RITM<br>sys_id>)                    | {"result":[{"state":"requested"}]}                                   |
| GET    | /api/now/table/sc_task                             | sysparm_query<br>(request_item=<RITM<br>sys_id>)                   | {"result":{"number":"SCTASK0010001","state":"Open"}}                 |

## 8. Authentication

- Users sign in to the instance with their ServiceNow credentials (or the organisation's single sign-on). A session is created after login and access is controlled by roles and ACLs.
- Only the requester can view their own requests; only the assigned manager can act on an approval; only the IT Procurement group works on the procurement task; administrators maintain the flow.
- REST API calls use basic authentication or OAuth tokens with the same role-based access.

## 9. User Interface

*Provide screenshots or GIFs showcasing different UI features.*

[ Insert screenshot: Service Portal – Standard Laptop Request form ]

[ Insert screenshot: Manager approval request ]

[ Insert screenshot: My Requests page showing request status ]

## 10. Testing

- Testing strategy: functional testing of each step of the flow using the Flow Designer Test option, followed by end-to-end testing by submitting requests as an employee and approving as a manager (impersonating users).
- Tools: Flow Designer Test and Executions view, user impersonation, email logs and the request / task records.

| No.  | Test case                         | Expected result                                                 |
|------|-----------------------------------|-----------------------------------------------------------------|
| TC-1 | Employee submits a laptop request | Request is created and a confirmation email is sent             |
| TC-2 | Manager approves the request      | Flow continues and a procurement task is created                |
| TC-3 | Manager rejects the request       | Flow stops and the employee is notified                         |
| TC-4 | Procurement task is completed     | Request is closed and the employee is notified                  |
| TC-5 | Flow execution fails              | Error is logged in Executions and the administrator is notified |

## 11. Screenshots or Demo

*Provide screenshots or a link to a demo to showcase the application.*

*[ Insert screenshot: Flow Designer – laptop order flow ]*

*[ Insert screenshot: Flow execution (successful run) ]*

*[ Insert demo link (if available) ]*

## 12. Known Issues

- If the requester has no manager set, the approval cannot be assigned and the request stays pending.
- Email notifications depend on the outbound email configuration of the instance.
- Vendor purchase-order integration is optional and is not part of the base lab flow.
- A Personal Developer Instance may go to sleep when inactive and must be woken up before use.

## 13. Future Enhancements

- Auto-approve standard laptop requests that are within a defined cost limit.
- Integrate the vendor's API for automatic purchase orders and delivery tracking.
- Add a live status tracker and a dashboard of order volumes and turnaround time.
- Add reminders and escalation for pending approvals and a laptop refresh reminder.
- Provide a Virtual Agent / mobile option for placing laptop requests.